

# VINAY THAKUR

Full Stack Developer | Backend Developer

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## PROFESSIONAL SUMMARY

Full Stack Developer with a strong foundation in backend engineering, building end-to-end web applications and networked systems using React, TypeScript, Node.js, and C++. Experienced in developing RESTful APIs, maintaining database consistency, and engineering concurrent systems for reliable performance under high load.

## TECHNICAL SKILLS

**Languages:** C++, JavaScript (ES6+), TypeScript, Python, SQL

**Frontend:** React.js, Vite, Tailwind CSS, HTML5, CSS3

**Backend:** Node.js, Express.js, RESTful APIs

**Databases:** MongoDB, Mongoose, PostgreSQL

**Tools & Platforms:** Linux, Git, GitHub, Postman, Neovim

**Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Concurrency, Multithreading, System Design Fundamentals

## PROJECTS

### High-Performance HTTP Server & Reverse Proxy

08/2026 – Present

C++ • TCP/IP • HTTP/1.1 • Multithreading • Load Balancing

- **Built** a multithreaded HTTP/1.1 server in C++ from raw TCP sockets, implementing **socket creation**, connection handling, HTTP request parsing, routing, and persistent connections.
- **Implemented** a thread pool with a thread-safe task queue to handle concurrent client connections and improve request-processing efficiency under load.
- Implemented a reverse proxy that forwards client requests to backend servers and relays responses, with Round-Robin load balancing across multiple upstream servers.
- Benchmarked the server at **35.8K requests/sec** using wrk with 8 threads and 1,000 concurrent connections, validating performance under high concurrency.

### Flash Sale Platform

06/2026 – Present

TypeScript • Node.js • Express.js • MongoDB • React

- **Full-Stack Architecture:** Built an end-to-end flash sale platform featuring decoupled React workflows for administrative sale management and high-throughput product purchasing with JWT authentication.
- **Concurrency & ACID Safety:** Engineered MongoDB multi-document transactions with Mongoose sessions to guarantee strict stock consistency, implementing a custom retry queue to handle **TransientTransactionError** write conflicts under heavy contention.
- **Lifecycle Management:** Designed automated lifecycle logic to enforce single-active-sale constraints per product and atomically restore allocated inventory upon sale cancellation.
- **Stress Testing & Validation:** Benchmarked system reliability using **k6**, validating 100% inventory accuracy (zero race conditions or double-spending) across 99 simultaneous VUs while isolating request throughput and lock latency.

## EDUCATION

**Master of Computer Applications - Lovely Professional University**

08/2023 – 07/2025

CGPA: 7.67

**Bachelor of Computer Applications - Himachal Pradesh University**

06/2020 – 07/2023

CGPA: 7.91

## VIRTUAL EXPERIENCE

**Forge, Hewlett Packard Enterprise Software Engineering Job Simulation**

08/2026 – 08/2026

- Designed a proposal for a RESTful web service to manage employee data.
- Built a Java Spring Boot web application capable of handling HTTP requests and supporting JSON data uploads.
- Developed and executed unit tests to validate application functionality and assess performance.

## CERTIFICATIONS

**REST API (Intermediate) | HackerRank**

Retrieving and processing API data using parameters and pagination.

**Hewlett Packard Enterprise - Software Engineering Job Simulation | Forge**

RESTful web services, Java Spring Boot, HTTP, JSON, and unit testing.